

INCIDENT TRIAGE TRAINING MANUAL

Document Control: SOC-TM-2026-V3

Target Audience: University Tier-1 Security Operations Center (SOC) Analysts

Operational Objective: Operational Workflow for Multi-View Triage, and Threat Vector Identification via Dashboard Architecture `SOC_Dashboard_Design_with_Images162.html`.

Section 1: Study Objectives & Framework

1.1 Why are we doing this? (The Main Objective)

The primary objective of this user study is to evaluate how **Explainable AI (XAI)** impacts a Security Operations Center (SOC) analyst's efficiency, trust calibration, and accuracy during alert triage. In traditional security environments, machine learning models act as "black boxes"—they flag an anomaly but provide no reasoning, leading to analyst fatigue, skepticism, or blind reliance. By introducing local feature attributions, we aim to discover if revealing the model's inner reasoning helps analysts differentiate true malicious insider threats from harmless, out-of-profile anomalies faster and with higher confidence.

1.2 Why do we have two designs? (The Analytical Comparison)

To measure the true impact of explainability, we must isolate the variable of XAI by comparing it against a control environment:

- **Design A (The Control):** Replicates the industry standard. It provides the analyst with plenty of raw logs, filters, and timeline graphs. The analyst must manually parse through a high volume of contextual data (such as login times, HTTP URLs, and device connections) to infer the threat scenario on their own.
- **Design B (The Experimental XAI Suite):** Introduces a transparent interface. Instead of forcing the analyst to hunt blindly through raw logs, it presents localized mathematical weights (SHAP waterfall charts) that immediately expose the top features driving the anomaly score, allowing the analyst to verify the model's logic through an interactive AI assistant.

1.3 What do we want to compare at the end? (The Evaluation Metrics)

By comparing the performance results of the analyst cohorts across both designs, this study measures three core technical areas:

- **Triage Efficiency:** Does XAI significantly decrease the time required to accurately diagnose a complex insider threat event?
 - **Diagnostic Accuracy:** Does XAI prevent critical misclassifications (such as misidentifying an active IP exfiltration to WikiLeaks as a routine unauthorized workspace login)?
 - **Justification Quality & Confidence:** Does XAI improve the clarity of the analyst's diagnostic narrative and elevate their certainty when signing off on a case closure?
-

Section 2: Glossary of Interface Naming Conventions

To execute high-efficiency triage operations, analysts must maintain absolute fluency in the system's naming conventions. Both operational dashboard views utilize a standardized, code-verbatim **Action Badge (ab-) prefix** system within the telemetry grids to separate, color-code, and filter distinct infrastructure log sources:

- `ab-login` : **Action Badge** designating user authentication, access tracking, and system login events.
- `ab-http` : **Action Badge** designating web traffic, proxy destination requests, and network upload activity.
- `ab-email` : **Action Badge** for outbound corporate mail routing and attached data metadata.
- `ab-file` : **Action Badge** for local file system modifications, mass directory queries, and data operations.
- `ab-device` : **Action Badge** for hardware attachment states, registry enumeration, and removable storage connection/disconnection events.

Section 3: Incident Threat Scenario Matrix

Tier-1 Analysts must cross-reference observed anomalous telemetry against the following standardized threat matrix to accurately classify malicious intent:

Scenario Code & Title	Tactical Description	Key Alert Indicators & Badges
S1: Removable Storage Exfiltration / IP Loss	An insider downloading proprietary intellectual property and transferring it to unapproved physical media or external web upload portals.	• <code>ab-device</code> (Hardware connection/disconnection) • <code>ab-file</code> (Mass file manipulation/search) • <code>ab-http</code> (Unauthorized external uploads/WikiLeaks access) • <code>alert-indicator email-exfil</code> (Data staging)
S2: Active Competitor Solicitation / Corporate Fraud	An employee searching for corporate job opportunities, communicating with industry competitors, or harvesting data for a future employer.	• <code>alert-indicator job-site</code> (Job board access) • <code>ab-email</code> (Outbound non-corporate routing) • <code>ab-http</code> (Competitor website traffic)
S3: Host Malicious Misuse via Keylogging Agent	A compromised or malicious endpoint utilizing hidden software to capture user credentials, master keystrokes, or intercept credentials.	• Unauthorized system software downloads • System configuration changes • Mass automated outbound mail triggers

Scenario Code & Title	Tactical Description	Key Alert Indicators & Badges
S4: Unauthorized Target Workspace Hijacking	An adversary or lateral-moving insider logging into an executive or restricted administrative account from an unmapped station.	• <code>ab-1ogon</code> (Anomalous authentication/times) • Out-of-profile administrative access • Sudden external cloud/home email configurations
S5: Multi-Vector Overlapping Profile	A highly critical complex threat signature demonstrating concurrent indicators from multiple scenarios simultaneously.	• Overlapping triggers (e.g., simultaneous <code>job-site</code> traffic, USB uploads, and off-hours administrative logins).

Part 1: Design A – Legacy Analytical View

1.1 Architectural Overview

Design A implements a classic SIEM architecture prioritizing raw chronological telemetry inspection, static threshold matching, and manual relational log verification to uncover host anomalies.

1.2 Step-by-Step Triage Workflow

Step 1: Statistical Severity Analysis

- Locate the **Topbar View Selection Interface** and select the "Design A (Legacy View)" toggle button.
- Review the **Statistical Metric Badges** on the top right to extract the current operational limits:
 - **Anomaly Cutoff Line:** The threshold vector dictated by Bayesian Threshold Optimization.
 - **True Catch Rate (Recall) & Precision Vector:** Real-time performance percentages.
- Identify user details using the **Google Account Identity Integration** inside the primary header text block (`User 162 (ALT4812)`). Note the underlying analytical engine models: an LSTM Autoencoder processing a CERT Insider Threat Dataset Baseline across a Mon–Fri, 8:00 AM–6:00 PM schedule.

Step 2: Timeline and Safety Deviation Audit

- Analyze the **Date-wise Anomaly Score Timeline Chart** to identify chronological spikes crossing the threshold.
- Toggle the **Timeline Z-Score Switch** in the table card header to overlay statistical standard deviations.
- Navigate the **Detected Anomalies Grid** to review day-by-day telemetry, comparing the **Anomaly Magnitude** against the **Safety Deviation Headroom**. Classify the status using the **Statistical Severity Label Badges**: `Normal` (Green/df-normal), `Elevated` (Orange/df-elevated), `High` (Red/df-high), or `Critical` (Purple/df-critical).

Step 3: Contextual Log Investigation

- Inspect the detailed action table beneath the timeline. Filter by activity type using the system's color-coded **Action Badges**: `ab-logon`, `ab-http`, `ab-email`, `ab-file`, and `ab-device`. Look out for details that match your threat profiles, specifically noting target URLs (e.g., whistleblowing/leak sites), exfiltration indicators, or data staging flags.

Step 4: Case Justification and Closure

- Scroll to the **Case Closure Classification Interface** at the bottom of the card.
- Click the **Information Circle Trigger** (`i`) to display the Hover Matrix Panel mapping your observations against known threat vectors.
- Map your findings to one of the predefined **Threat Scenario Classifications** (`S1` through `S5`).
- Select your **Analyst Confidence Level** from the dropdown menu (Scale 1–5).
- Complete the **Analyst Case Justification Narrative** text area by providing a mandatory 3-line tactical diagnostic summary.
- Select your verified identity inside the **SOC Analyst ID Select Tool**, verify session timer metrics via the elapsed display, and click **Export Forensic Evidence Package** to download the structured mitigation package.

Mock Triage Walkthrough (Design A)

Scenario Context: You log into your shift to audit the active dashboard environment loaded for User 162.

1. **Observation:** You look at the **Date-wise Anomaly Score Timeline Chart** and observe a massive, isolated chronological spike on the date **October 25**.
2. **Telemetry Audit:** You scroll to the **Detected Anomalies Grid**. The entry for **October 25** is highlighted with a `Statistical Severity Label` of **Critical**. The logged `Anomaly Magnitude` far exceeds the `Anomaly Cutoff Line`, dropping the `Safety Deviation Headroom` to absolute zero.
3. **Log Analysis:** You filter the raw log table below to exclusively isolate the events of **October 25**.
 - At 01:15 AM (outside the user's standard 8:00 AM–6:00 PM workflow baseline), an `ab-logon` event flags a successful interactive remote login session from a completely unmapped internal station.
 - At 01:22 AM, a surge of purple `ab-file` badges appears with details indicating an uncharacteristic bulk query and local directory staging of corporate source repositories.
 - At 01:38 AM, a green `ab-http` badge registers web traffic directed to a restricted upload node on **WikiLeaks.org**, instantly triggering an `alert-indicator email-exfil` data staging alert.
4. **Triage Assessment:** Scanning the destination network logs to an explicit whistleblowing leak portal confirms the core objective is corporate intellectual property theft. This dictates classification under **S1: Removable Storage Exfiltration / IP Loss**, overriding the initial remote logon vector flag (\$S4\$).
5. **Disposition:** You open the **Case Closure Classification Interface**. You select **S1 - Removable Storage Exfiltration / IP Loss** from the dropdown. You assign an **Analyst Confidence Level of 5 (Highly Confident)**. In the narrative profile, input:
 - *Line 1: Critical baseline anomaly observed on October 25 at 01:15 AM involving unauthorized out-of-*

hours interactive login.

- Line 2: Staged corporate source data was actively exfiltrated via an outbound HTTP upload stream to WikiLeaks.org.
- Line 3: Confirmed malicious IP exfiltration (S1); endpoint isolated, credentials revoked, and legal/incident response teams notified.

6. **Closure:** You select your Analyst ID, verify the session timer, and click **Export Forensic Evidence Package**.

PART 2: DESIGN B – EXPLAINABLE AI (XAI) SUITE VIEW

2.1 Interface Overview

Design B shifts the triage paradigm to Explainable AI (XAI) analytics. It visualizes localized mathematical contributions to individual anomaly metrics, utilizing SHAP (SHapley Additive exPlanations) waterfall plots and population beeswarm arrays to explain *why* the machine learning model flagged an event.

2.2 Step-by-Step Triage Workflow

1. **Active Mode Activation:** Engage the **Architecture Selector Group** and choose "Design B (XAI Suite)". Verify that the main card body refreshes to load the **SHAP Per-Day Feature Contribution Grid** alongside the population map module.
2. **Local Feature Attribution Analysis:** Examine the **SHAP Waterfall Grid Thumbnails**. Each image block displays a decomposition of a single day's anomaly score. Hover over the **Waterfall Information Tooltips** (ⓘ) to refresh your understanding of the XAI calculations ($E(x)$ Expected Value, $f(x)$ Actual Output, Red positive attribution bars, Blue negative attribution bars). Cross-reference active plots against the **Scenario-Specific Feature Checklist tooltips** to detect target signatures:
 - Check for **S1 signatures** by hunting for prominent red bars linked to `ab-device` (thumb drive usage), night activities, connect/disconnect logs, or unauthorized external domain web uploads via `ab-http` (such as WikiLeaks).
 - Check for **S2 signatures** by validating red forces caused by active `alert-indicator` `job-site` access, competitor `ab-email` logs, and localized thumb drive storage patterns.
 - Check for **S3 signatures** by identifying unique red attribution weights driven by malicious software or keylogger downloads, unauthorized software logins, and mass outbound e-mail storms.
 - Check for **S4 signatures** by auditing red forces tied to unauthorized remote `ab-logon` sessions, rapid localized file searches, and anomalous home email network access.
 - Check for **S5 signatures** by spotting a dense, multi-vector layer of overlapping red bars where device, web, logon, and email metrics all trigger major positive attribution forces simultaneously.
3. **Interactive Local Assistant Calibration:** Click the contextual utility buttons inside the **Local Waterfall Interpretation Assistant Row** to refine your review, toggling between `Core Interpretation`, `How to Read Bars`, `Red vs Blue Forces`, and `Next Steps for Triage` to dynamically update the XAI guide box.
4. **Context-Stack Querying via Gemini LLM Engine:** Click on a specific **Thumbnail Card Frame** to mount it to the active investigative workspace. Observe that the card structural border changes color to violet

(**selected-for-chat**), and the **Target Context Stack Bar** dynamically locks onto the selected target profile. Use the **LLM Chatbot Field** at the base of the framework to audit feature weights. Input questions regarding anomalies or specific data vector spikes, then click **Send Query to AI Master** to generate an automated contextual breakdown from the integrated model.

5. **Global Population Calibration:** Scroll to the **Global Population Model Beeswarm Map**. Assess how the current subject's local feature behaviors map relative to the wider cluster database, identifying if the data point centers within a high-density, malicious insider pattern cluster.
6. **Verdict Entry and Forensic Export:** Once intent is established through XAI analysis, return to the **Detected Anomalies Grid**. Execute the standard closure routine: categorize the case classification (**S1** to **S5**), set your confidence score, select your **SOC Analyst ID**, fill in the 3-line narrative justification, and generate the package using the **Export Forensic Evidence Package** button.

Mock Triage Walkthrough (Design B)

Scenario Context: You are actively triaging User 162 via Design B (XAI Suite) in the provided system panel. The date **October 25** has generated an active alert.

1. **Initial Image Audit:** You look at the **SHAP Waterfall Grid** for **October 25**. The base expected value $E(x)$ remains low, but the final value $f(x)$ is pushed highly positive by anomalous features.
2. **Attribution Signature Detection:** You look at the directional bars and observe zero blue bars pulling the score down, and a cluster of heavy red bars extending to the right.
 - The primary red driver bar is labeled with the feature **ab-http** (specifically tracking external upload data directions to WikiLeaks.org).
 - The secondary red driver bar is labeled with the feature **ab-file** (tracking mass file staging operations).
 - The tertiary red driver bar is labeled with the feature **ab-logon** (tracking out-of-hours remote interactive authentications).
3. **Scenario Mapping:** You verify these features against the **Scenario-Specific Feature Checklist** tooltips. The mathematical presence of unauthorized external HTTP uploads and mass file actions precisely matches **S1: Removable Storage Exfiltration / IP Loss**.
4. **AI Engine Deep-Dive:** You click on the **October 25** Waterfall Thumbnail; the card lights up as **selected-for-chat** and the target context stack bar locks onto **October 25**. In the chat window, you type: *"Analyze the feature weights for ab-http and ab-file on October 25 and evaluate data exfiltration risk."*
5. **LLM Verdict Evaluation:** The integrated Gemini engine returns an analysis: *"The extreme positive SHAP attribution for ab-http maps directly to data transmission spikes toward a public leak platform. Paired with the high red attribution for ab-file (data repository staging) and the late-night ab-logon vector, this indicates a deliberate, multi-stage data breach. This confirms the S1 exfiltration profile."*
6. **Global Verification:** You scroll down to check the **Global Population Model Beeswarm Map**. The coordinate for **October 25** sits completely isolated on the far right extreme of the graph, clustering strictly with historical malicious control baselines.
7. **Case Resolution:** Confirmed breach. You select **S1 - Removable Storage Exfiltration / IP Loss** from the closure classification dropdown, mark your confidence as **5 (Highly Confident)**, insert your 3-line investigative narrative confirming the WikiLeaks exfiltration target and baseline overrides, and click **Export Forensic Evidence Package**.